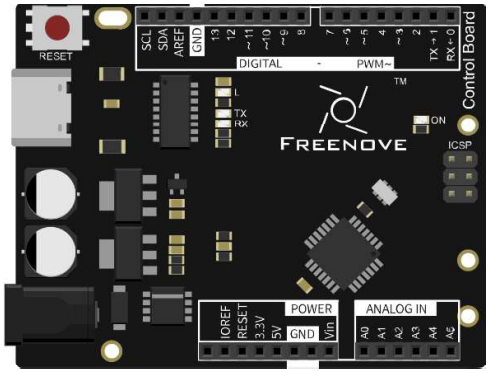
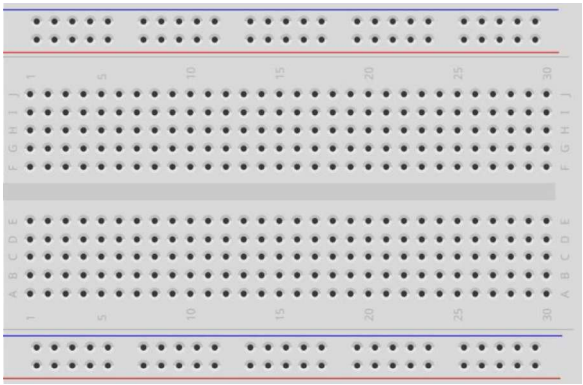
Project 8.2 Multicolored LED

In the previous section, we have finished controlling the RGB LED to emit light with different color and brightness through three potentiometers. Now, we will try to make RGB LED emit multicolored lights automatically.

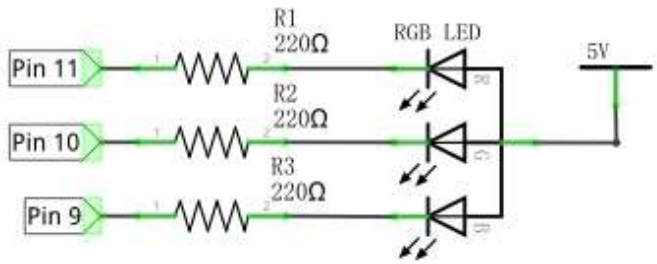
Component List

Control board x1 	Breadboard x1 
USB cable x1	RGB LED x1	Resistor 220Ω x3
Jumper M/M x4	

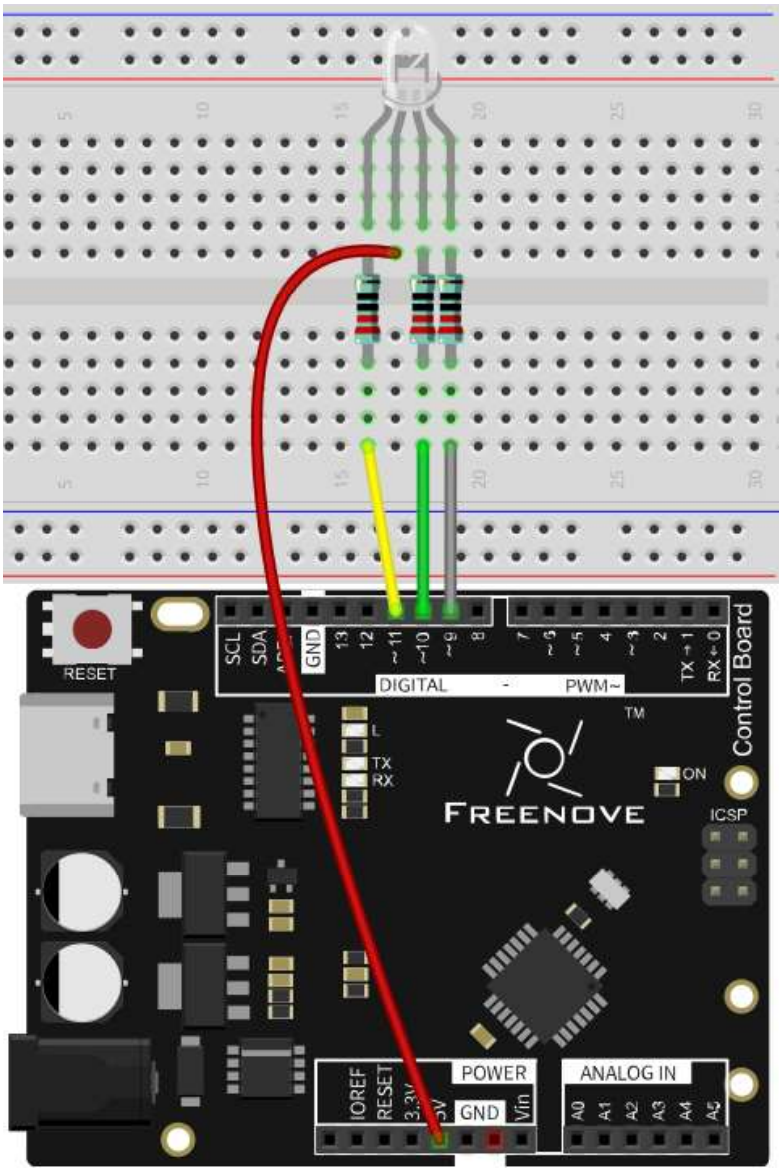
Circuit

Use pin 9, 10, 11 of the control board to control RGB LED.

Schematic diagram



Hardware connection



Sketch

Sketch 8.2.1

Now, write code to generate three random numbers, and convert them into PWM duty cycle to control the three LEDs of RGB LED.

```
// set pin numbers:
int ledPinR = 11;   // the number of the LED R pin
int ledPinG = 10;   // the number of the LED G pin
int ledPinB = 9;    // the number of the LED B pin

void setup() {
  // initialize the LED pin as an output:
  pinMode(ledPinR, OUTPUT);
  pinMode(ledPinG, OUTPUT);
  pinMode(ledPinB, OUTPUT);
}

void loop() {
  // Generate three random numbers between 0-255 as the output PWM duty cycle of ledPin
  rgbLedDisplay(random(256), random(256), random(256));
  delay(500);
}

void rgbLedDisplay(int red, int green, int blue) {
  // Set three ledPin to output the PWM duty cycle
  analogWrite(ledPinR, constrain(red, 0, 255));
  analogWrite(ledPinG, constrain(green, 0, 255));
  analogWrite(ledPinB, constrain(blue, 0, 255));
}
```

In the code, we create three random numbers, and convert them into PWM duty cycle to control the three LEDs of RGB LED to emit light with different brightness. At regular intervals, a new random number will be created, so RGB LED will start flashing light with different colors and brightness.

random(min, max)

random (min, max) function is used to generate random number, and it will return a random value in the range (min, Max-1).

You can also use random (max) function, the function set the minimum value into 0 by default, and returns a random value in the range (0, Max-1).

Verify and upload the code, and RGB LED starts flashing with different colors and brightness.